From requirement to verified implementation

A constraint-driven path through planning, execution, verification, evidence, acceptance, and recovery.

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> Claims change state only when their evidence supports the transition.

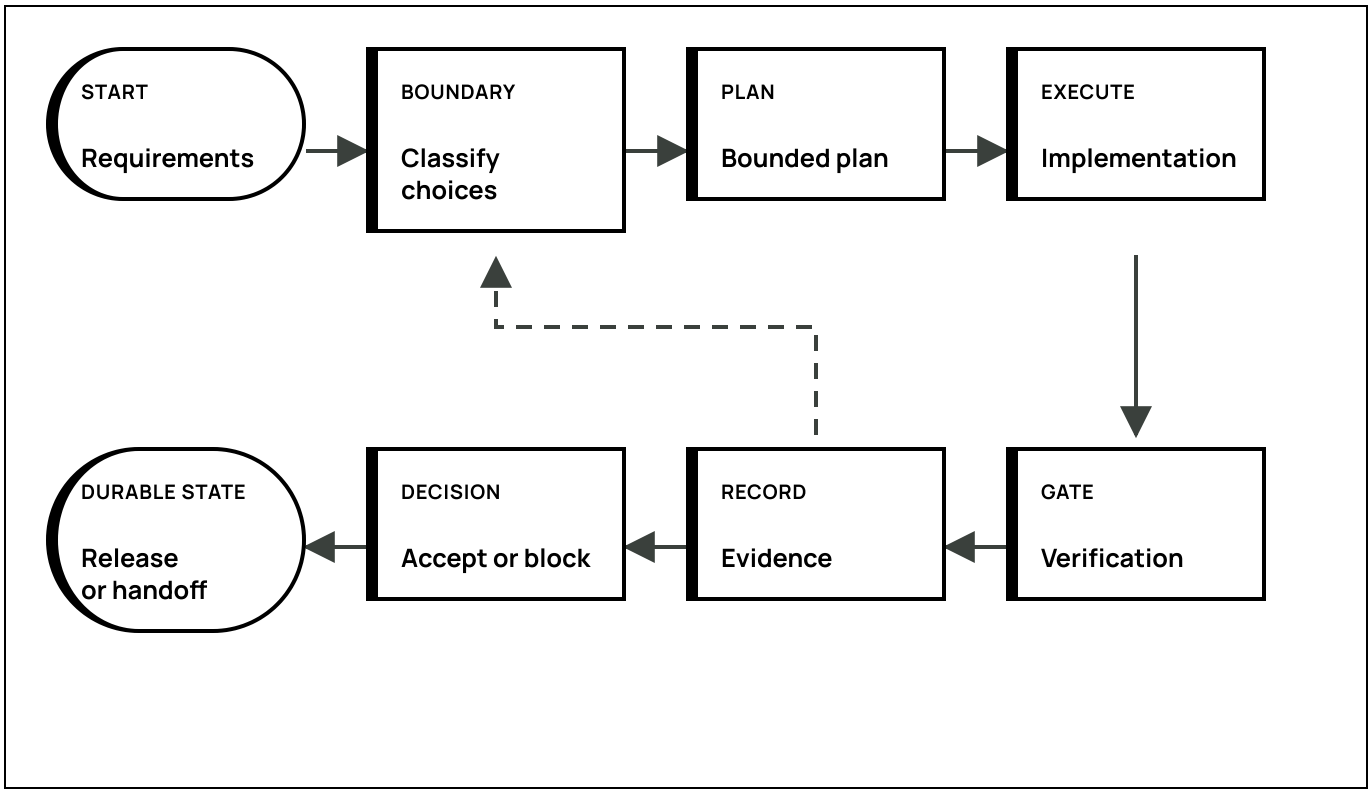
D02 · ENGINEERING LIFECYCLE

The actual path includes rejection and recovery

A requirement reaches acceptance only through classified choices, bounded execution, proportionate verification, and durable evidence.

D02 — Requirements to verified implementation

The main path is solid. Failed verification returns work to correction; an invalid or unsupported choice stops before implementation.



1. Requirements lead to decision classification, then a bounded plan and implementation.
2. Implementation must pass proportionate verification before evidence can support acceptance.
3. Failed verification returns to classification and planning for correction.
4. Acceptance produces a releasable result; a blocker produces a durable handoff instead of a false success.

REQUIREMENTS TO VERIFIED IMPLEMENTATION

The actual work path includes rejection and recovery

The lifecycle is not a success-only funnel. Invalid transitions stop before execution; failed checks return work to correction.

01	**Define.**	State the requirement, constraints, source authority, and measurable acceptance criteria.
02	**Classify.**	Separate predetermined decisions, delegated choices, required approvals, and prohibited actions.
03	**Plan.**	Select the smallest process that preserves correctness, traceability, and continuity.
04	**Execute.**	Implement in bounded slices and preserve evidence at coherent recovery points.
05	**Verify.**	Run applicable build, structural, browser, accessibility, and print checks.
06	**Accept or recover.**	Record pass, fail, blocked, or not run; resolve failures or hand off exact blockers.

CONTROL POINTS

Verification changes what may happen next

Human authority, machine-enforced rules, and evidence gates remain distinct.

BEFORE EXECUTION

Authorization gate

The proposed transition must be legal and inside the granted scope.

`Documented contract`

DURING EXECUTION

Evidence gate

Important claims and completed work must have durable, inspectable support.

`Documented contract`

BEFORE COMPLETION

Verification gate

Completion requires the checks appropriate to the risk; unavailable checks stay visible.

`Documented contract`